

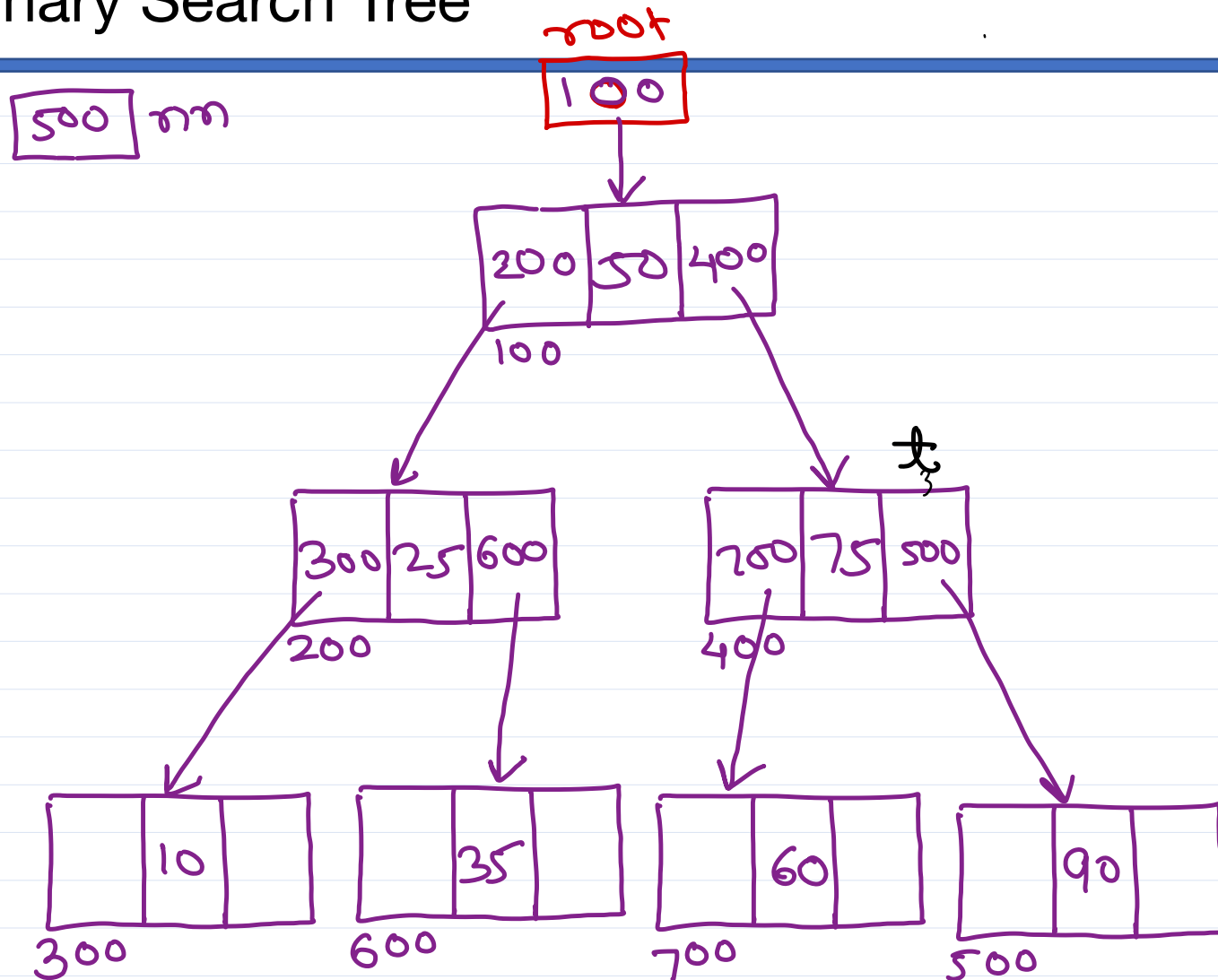


Sunbeam Infotech

Exploring new ideas, Reaching new heights!



Binary Search Tree



add()

✓ 50

✓ 25

✓ 10

✓ 75

✓ 90

35

60

```

class Node {
    int val;
    Node left;
    Node right;
    ...
}
  
```

```

class Tree {
    Node root;
    Tree() {
        root = null;
    }
    void add(int v) {
        //...
    }
}
  
```

```

    void preorder() {
        //...
    }
}
  
```



```
void add (int v) {
```

```
    Node nn = new Node (v);
```

```
    if (root == null)
```

```
        root = nn;
```

```
    else {
```

```
        Node trav = root;
```

```
        while (true) {
```

```
            if (v < trav.val) {
```

```
                if (trav.left == null) {
```

```
                    trav.left = nn;
```

```
                    break;
```

```
                }  
                trav = trav.left;
```

```
            }  
            else {
```

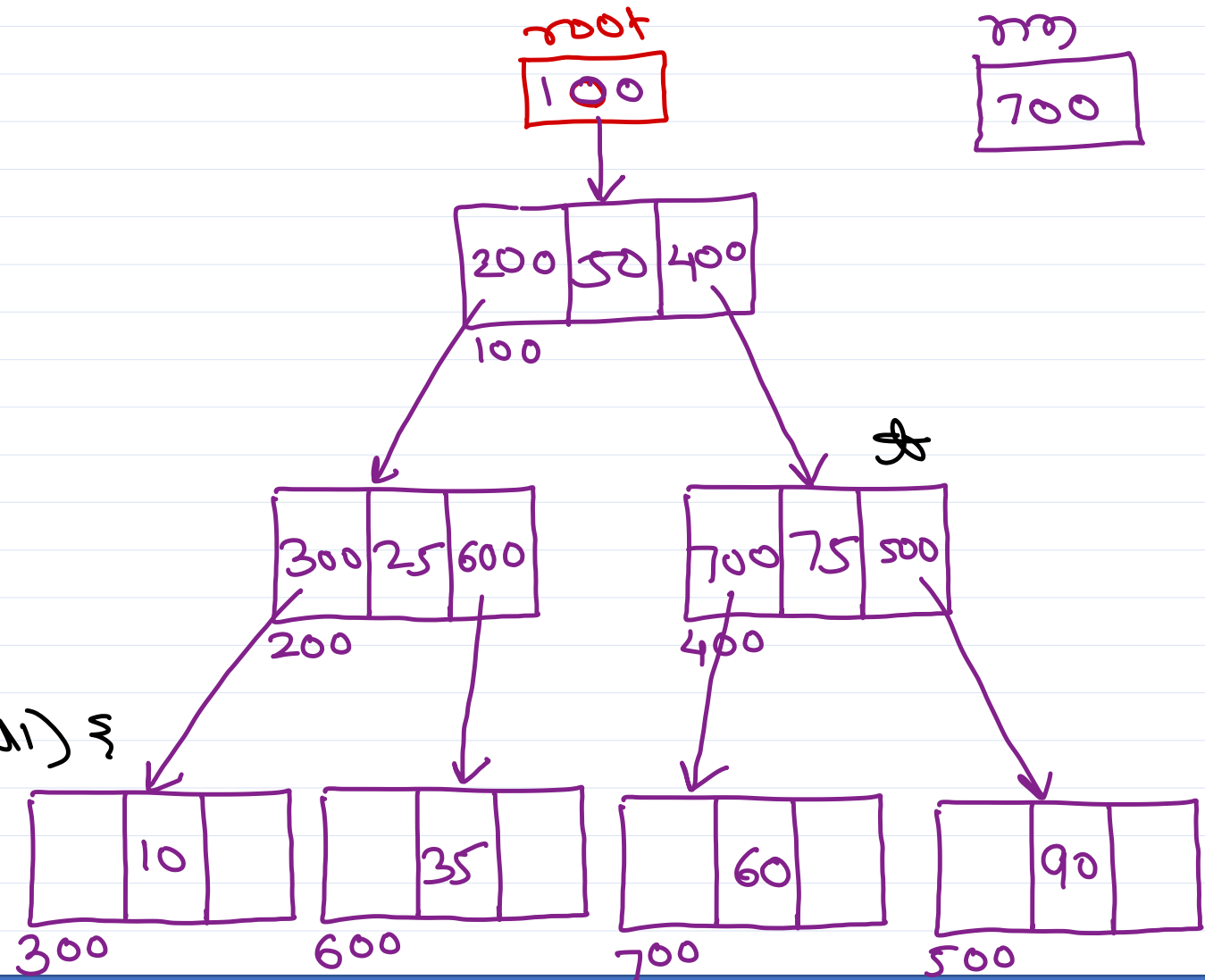
```
                if (trav.right == null) {
```

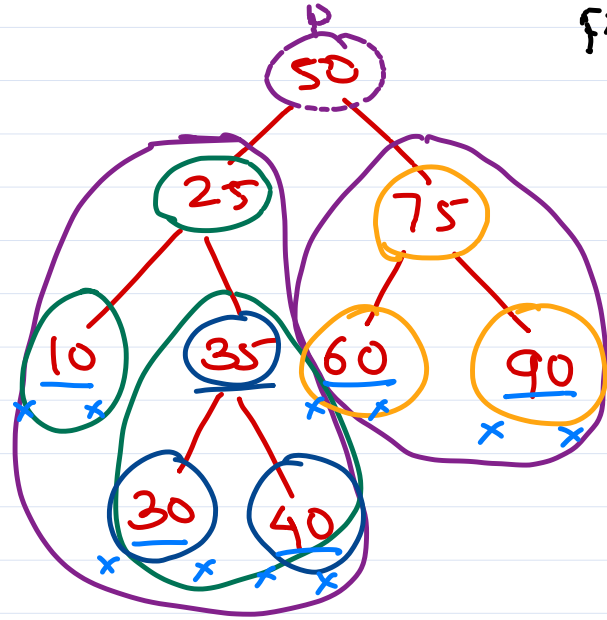
```
                    trav.right = nn;
```

```
                    break;
```

```
                }  
                trav = trav.right;
```

```
        }  
    }  
}
```





Preorder: P L R

50 25 10 35 30 40

75 60 90

```

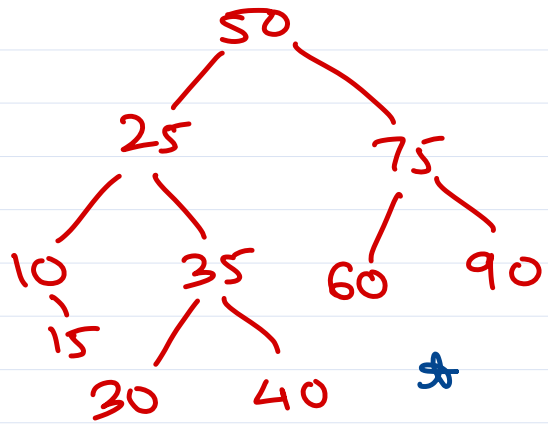
void preorder(Node trav) {
    if (trav == null)
        return;
    cout << trav->data;
    preorder(trav->left);
    preorder(trav->right);
}

```

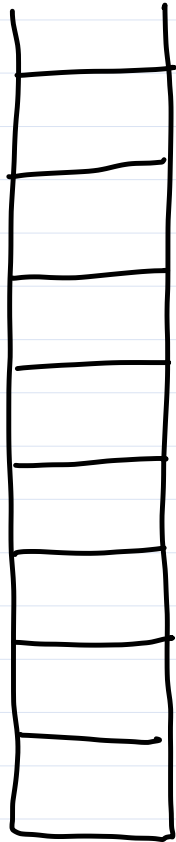
3



Preorder without recursion



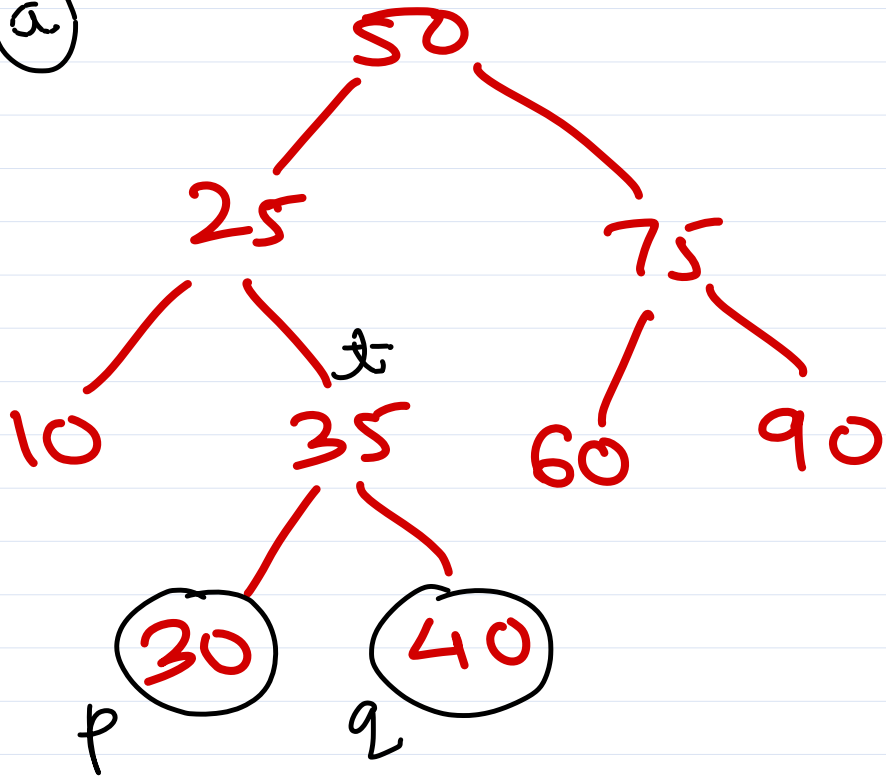
50 25 10 15
35 30 40 75
60 90



```
stack<Node> s = new Stack();  
Node trav = root;  
while (trav != null || !s.isEmpty()) {  
    while (trav != null) {  
        print(trav.val);  
        if (trav.right != null)  
            s.push(trav.right);  
        trav = trav.left;  
    }  
    if (!s.isEmpty())  
        trav = s.pop();  
}
```



(a)



```
Node ancestor(trav, p, q) {  
    while(trav != null) {  
        if(p.val < trav.val && q.val < trav.val)  
            trav = trav.left;  
        else  
            if(p.val > trav.val && q.val > trav.val)  
                trav = trav.right;  
            else  
                return trav;  
        }  
    }  
    return null;  
}
```

}





Thank you!

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